GY462 — Real Estate (RE) Finance

Lecture 3

1

Operating CF Proforma
—*Typical Line Items*

- **Operating (all years)**

Potential Gross Income (Rent*SF)	=	PGI
– Vacancy allowance (=Vac Rate*PGI)	=	– V
+ Other Income	=	+ OI
– Operating expenses	=	– OE
= Net Operating Income	=	NOI
– Capital Improvement Expenditures	=	– CI
(O) Property Before-Tax Cash Flow	=	PBTCF

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Reversion CF Proforma
—*Typical Line Items*

- **Reversion**
  (last year & years of partial sales)

Property value at time of sale	=	PGI
– Selling Expenses	=	– SE
(R) Property Before-tax Cash Flow	=	PBTCF

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Resale Value
– *How to Determine?*

- **Consider typical 10 year DCF valuation**
  - Projected reversion typically > 1/3 of PV
- **Widespread resale forecasting method**
  - Divide year 11 NOI by 'going out' cap rate
- **Going out cap rate relative to going in cap rate?**
  - Usually going out cap ≥ going in cap
  - Why?

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'Focus point 1'

Going out cap ≥ going in cap?!
How do I make money (i.e., sell at a premium)?

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Where We Are…

1. **Investment Analysis**
   - Valuation of Income Producing Properties
     - DCF Method in Real Estate Context
     - Shortcut Methods: Cap Rate and GIM
     - Choice of Discount Rate
   - Pro Forma Basics
   - **Investment Decision**
     - **Measures of Investment Performance**
     - **NPV and IRR Investment Rules**
   - Risk Assessment Methods

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Measures of Performance — *Overview*

- **Measures of investment performance using ratios**
  - **Price per square foot**
  - **Cap rate**
  - **Equity dividend rate**
  - **Debt coverage ratio**

  Short-cut methods – *in brief*
- **Measures of investment performance based on CF projections**
  - **NPV**
  - **IRR**

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NPV Investment Rule

- **Maximize NPV across all mutually exclusive alternatives with NPV≥0**
- **Zero NPV deals are ok and expected!**

NPV(Buyer) = Market Value – Price
NPV(Seller) = Price – Market Value

⇨ NPV(Buyer) = – NPV(Seller)

Since deal only if NPV(Buyer) = NPV(Seller)
Thus NPV(Buyer)=NPV(Seller)=0

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IRR Investment Rule

- **Question: What discount rate will cause expected future cash flows to be worth the given price?**
- **Rule:**
  1. **Maximize difference between project's expected IRR and required return (i.e. benchmark rate + RP)**
  2. **Never do deal with expected IRR less than required return ("hurdle rate")**

IRR Computations – *Details*

- **Consider 10 year projection**
- **First 6 years: existing lease (step-up)**
- **Next 4 years: expect new lease, with uncertainty but higher value**
- **Expected sale price in year 10: 20**
- **I am being asked to pay 13.453 for this property**

⇨ **IRR of investment?**

Year	CF
2001	1.0
2002	1.0
2003	1.0
2004	1.5
2005	1.5
2006	1.5
2007	2
2008	2
2009	2
2010	2

How to compute IRR with calculator

IRR: Discount rate such that PV of future cash inflows = initial cash outflow

Financial Calculator Solution (TI BA II PLUS)

CF	(Select cash flow worksheet)		
CF0=	–13.45	ENTER	(Enter initial cash payment, PV)
C01=	1	ENTER	(enter 1st CF)
F01=	3	ENTER	(enter n. of periods of 1st CF)
C02 =	1.50	ENTER	(enter 2nd CF)
F02=	3	ENTER	(enter n. of periods of 2nd CF)
C03=	2	ENTER	(enter 3rd cash flow)
F03=	3	ENTER	(enter n. of periods of 3rd CF)
C04=	22	ENTER	(enter last period CF+sale value)
F04=	1	ENTER	(enter n. of periods of last CF)
IRR CPT	=13.13		(computes internal rate of return)

Deal too Good to be True?!...

- Land cost at historical value
- **Mismatched growth rates between rental income and expenses**
- Missing operating expenses
- **Failure to consider rental concessions and effective rents**

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Deal too Good to be True?!... *Continued*

- Vacancy rates too low
- **Use of low going-out capitalization rates**
- Underestimation of sales and other reversion costs

Where We Are...

1. **Investment Analysis**
   - Valuation of Income Producing Properties
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   - Pro Forma Basics
   - Investment Decision
     - Measures of Investment Performance
     - NPV and IRR Investment Rules
   - **Risk Assessment Methods**

Risk Assessment

- **Investment performance measures are only comparable if risk is the same**
  - **What risks should I consider/bear in mind?**
  - **How do I measure risk and assess it effect on my investment?**

What Kind or Risks?

- **Different investments are exposed to different types of risks to different degrees**
- **<u>Types of risk</u>: *Business risk, financial risk, liquidity risk, inflation risk, management risk, interest rate risk, legislative risk, environmental risk***

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Risk Assessment Methods — *Overview*

- **Many ways of assessing risk, present three important ones:**
  1. **Due Diligence**
  2. **Partitioning the IRR/PV**
  3. **Sensitivity Analysis**

Due Diligence

- **Systematic process used in the RE investment community to assess suitability of investment prior to acquisition**
  - **Question: Is investment risk suitable given a set of investment objectives?**
  - **Important when firm makes investment on behalf of other investors** (accountability!)

Partitioning the IRR/PV

- **Determine relative contributions of**
  - **CFs from operation**
  - **CFs from sale of investment** (less certain!)
- **Office building example**
  - **PV operation: 30% of total PV**
  - **PV sale: 70% of total PV**

Sensitivity Analysis

- **Objective: Quantitative perspective to help rank investment alternatives**
- **Evaluate IRR under various scenarios**
- **Assign weights to each scenario…**

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Sensitivity Analysis — *Continued*

- **Compute expected return**
  - $IRR_E = Pr_1\, IRR_1 + Pr_2\, IRR_2 + Pr_3\, IRR_3 + \ldots$
- **Compute variance, standard deviation, coefficient of variation (CoV)**
  - $VAR = Pr_1\,(IRR_1 - IRR_E)^2 + Pr_2\,(IRR_2 - IRR_E)^2 + \ldots$
  - $SD = VAR^{1/2}$
  - $CoV = SD(IRR) \div E(Mean(IRR))$

'Focus point 2'

Scenario analysis:
Are the expressions clear?
What elements of the cash flow would you 'flex'?

Focus of this lecture and outlook…

- **Rest of today and next week we will cover**
  - **Effects of financial leverage on return/risk**
  - **Concept of incremental cost of debt**
  - **Assess advantages and disadvantages of borrowing**

…then development project and PE / funds

Chapter 2 — *Overview*

2. **Financial Leverage**
   - **Leverage and Returns**
     - **Example**
     - **Incremental Cost of Debt**
   - **Leverage and Yield/Growth**
   - **Leverage and Risk**
   - **Other Considerations**

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Why Use Debt?

- Lack of equity capital
- Profit from tax deductibility of mortgage interest
- Realize benefits from <u>financial leverage</u> when borrowing rate is lower than expected rate of return on investment
- Change risk characteristics of project

Some Terminology

- Asset Value (V)
  = Debt Value (D) + Equity Value (E)
- Loan to Value Ratio (LTV)
  = D / V
- Leverage Ratio (LR)
  = V / E = V / (V – D) = 1 / (1– D / V)
  = 1 / (1 – LTV)

Example (based on US tax rules)

<u>Purchase Price</u>	Building value	85
	Land value	15
	Total value	100
<u>Income Assumptions</u>	NOI (p.a.)	12
	Income tax rate	28%
	Depreciation	31.5 years *(straight line)*
	Holding period	5 years
	Resale price	100
<u>Loan Assumptions</u>	Amount	80
	Interest	10%
	Term	Interest only (non amortizing)

⇨ LTV = 80%
⇨ LR = 1 / (1 - 0.8) = 5

Cash Flows with Debt

- $BTCF_O = 12 - 8$
- Taxable income = 12 – 8 – 85/31.5 = 1.302
- $ATCF_O = 4 - (1.302 \times 0.28) = 3.636$
- $BTCF_S = 100 - 80 = 20$
- Adjusted basis = 100 – 5 × (85/31.5)
- $Taxes_S = [100 - (100 - 5 \times (85/31.5))] \times 0.28 = 3.778$
- $ATCF_S = 20 - 3.778 = 16.222$

Year	*0*	*1-4*	*5*	*IRR_{Debt}*
BTCF	-20	4	24	*20.0%*
ATCF	-20	3.636	19.858	*15.4%*

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Cash Flows with Debt – *Some Remarks*

- $BTCF_O$ = Cash Flow – Interest payment
  - Only interests are deductible, not principal payment
- Taxable income = $BTCF_O$ – Initial Value of Building / Depreciation Allowance
- $ATCF_O$ = $BTCF_O$ – (Taxable income × Tax rate)
- Important: only depreciating initial value of the building, not the land
  - Only 85/31.5, not 100/31.5
  - Depreciation is for physical investment, land is excluded

Cash Flows with Debt – *More Remarks*

- $BTCF_S$ = Resale value – outstanding loan
  - Here outstanding loan is complete loan (interest only)
- How much taxes? Consider capital gains
  - If no depreciation they are zero: 100 to buy and 100 from sale. However…
- Adjusted capital basis = Original cost – n. of years × (yearly depreciation of capital)
- Taxes $_S$ = [Resale Value – Adjusted Capital (i.e. Depreciated value)] × Tax Rate
- $ATCF_S$ = Resale Value – Taxes

Cash Flows with No Debt

- $BTCF_O$ = 12
- Taxable income = 12 – 85/31.5 = 9.302
- $ATCF_O$ = 12 – 9.302 × 0.28 = 9.396
- $BTCF_S$ = 100
- Taxes $_S$ = [100 – (100 – 5 × (85/31.5))]×0.28 = 3.778
- $ATCF_S$ = 100 – 3.778 = 96.222

	0	1-4	5	IRR	IRR_{Debt}
BTCF	-100	12	112	12.0	*20.0*
ATCF	-100	9.396	105.618	8.76	*15.4*

'Focus point 3'

IRR with and without leverage:
Can you work out the four IRRs above?
What do you notice when you compare them?

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Break… ☺

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Debt as a Return Magnifier

- **Observation: BTIRR $_{\text{With Debt}}$ >> BTIRR $_{\text{No Debt}}$**
- **Why?**
  - **Unlevered BTIRR** *(= 12%)* **greater than cost of debt** *(= 10%)*
  - **Positive leverage!**
  - ⇨ **What is the precise relationship between these measures?**

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Debt as a Return Magnifier — *Formalizing, Before Tax*

- **Accounting Identities**
  - *Asset = Liabilities + Owner's Equity*
  - *Asset CF = Debt CF (Mortg.Paym.) + Equity CF*
- **Let** $r_D = DCF / D$
  $r_E = ECF / E$
  $r_P = CF / V$

Weighted Average Cost of Capital (WACC):

$$r_P = (LTV)r_D + (1 - LTV)r_E$$

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Computation WACC - *Remarks*

- **WACC** $r_P = (LTV)r_D + (1 - LTV)r_E$
- **Which can also be re-written as…**

$$r_E = r_P + (r_P - r_D)\,{}^{D}\!/_{E}$$

- **What is clear from this algebra?**
  - **As long as return on unleveraged investment is above cost of debt…**
  - **The return on equity is magnified by borrowing**

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Computation WACC - *Example*

- **Accounting Identities**
  - *Asset (100) = Liabilities (80) + Equity (20)*
  - *Asset CF = 12 = 8 + 4*
- **Let** $r_D = 8 / 80 = 10\%$
  $r_E = 4 / 20 = 20\%$
  $r_P = 12 / 100 = 12\%$

Weighted Average Cost of Capital:

$$20\% = 12\% + (12\% - 10\%) \times 8/2$$

Positive Leverage After Tax?

- **In example:**
  - **Cost of debt** *(10%)* > **ATIRR no debt** *(8.76%, after tax)*
  - Nevertheless:
    ATIRR 80% debt *(15.4%)* > **ATIRR no debt** *(8.76%)*
- **Why?**

Positive Leverage After Tax - *The Explanation*

- **Reason: Interest is tax deductible**
- **After tax cost of debt:** 0.1*(1 – 0.28) = 7.2% (< 8.76%)
  - Can guess that return on equity is bigger if use loan…
- **Consider WACC formula and reformulate to get return on equity:**

$r_E = r_P + (r_P - r_D)\ (D/E)$

r_E = **8.76 + (8.76–7.2) × (80/20) = 15 ≠ 15.4**

⇨ **Not identical to DCF results. Why?**

Positive Leverage After Tax —*Continued*

- **Debt to equity ratio increases over holding period!**
  - **Initial D/E = 80 ÷ 20 = 4**
  - **D/E_S = 80 ÷ 16.222 = 4.93**
- **Use WACC formula with initial D/E as approximation!**

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'Focus point 4'

**WACC vs. IRR calculations:
Why is WACC an approximation?**

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Break-Even Interest Rate *(BEIR)*

- **When is leverage favorable?**
  - **Rate such that** $ATIRR_D = ATIRR_P$

    ⇨ **Break Even Interest Rate** *(BEIR)*
- **Definition: Maximum interest rate that could be paid before the leverage becomes unfavorable**
- **Can approximate** *(see above)* $BTIRR_D\,(1 - t) = ATIRR_D$

  ⇨ $BTIRR_D = ATIRR_D / (1 - t)$
- **Now:** $BEIR \Leftrightarrow ATIRR_D = ATIRR_P$
- **Implying** $BEIR = ATIRR_P/(1 - t) = 8.76/0.72 = 12.17$

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Incremental Cost of Debt

- **Example: 80% loan at 10%**
- **Alternative: 85% loan at 10.25%**
- **Worth it?**
  - **Extra 5 for extra 0.25% <u>on whole loan</u>**
  - **85 * 10.25% – 80 * 10% = +0.7125** *(difference per year)*
  - **Incremental cost of additional 5:**
    0.7125 / 5 = 14.25% > ????

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Leverage and Yield

- **WACC formula can be applied to any additive component of total return** r
- $r = y + g$; $r = r_f + RP$
- **Components**
  - **income yield** y
  - **growth** g *(property value appreciation)*
  - **risk premium** RP

⇨ **…**

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Leverage and Yield

- **Apply WACC formula to income yield:**

$$y_P = (LTV)\, y_D + (1 - LTV)\, y_E$$

- **Rearranging:**

$$y_E = \frac{(y_P - LTV \times y_D)}{(1 - LTV)}$$

  - **Equity cash yield, or <u>cash-on-cash yield</u>**

Leverage and Growth

- **Again from WACC**

$$g_P = (LTV)\, g_D + (1 - LTV)\, g_E \quad \Rightarrow g_E = \frac{(g_P - LTV \times g_D)}{(1 - LTV)}$$

- **What is g_D?**

$r_D = g_D + y_D$

$y_D = CF_D / V_D$

so g_D is change in value of debt (<u>amortization</u>)

<u>'Focus point 5'</u>

**Leverage, cash-on-cash and growth yield:
are these concepts clear?
And why/how are they helpful?**

Example: Japanese Property

- **Buy £100 million of Japanese RE at cap rate of 7%**
- **Assume falling net rents ~ -2% p.a. (in 2006) ⇨ Justification?**
  - **No noteworthy (nominal) economic growth over last 13 yrs**
  - **Real estate supply growth ~+2% p.a.**

Year	2006	2007	2008	2009
NOI	7	6.86	6.72	6.59

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Example: Japanese Property — *Continued*

- **Further assumptions**
  - **LTV = 85%**
  - **Interest rate = 4% (!)**
  - **Non amortising loan**
    ⇨ interest payments p.a. = 3.4

⇨ **Cash-on-cash yield?**

Japanese Property *Cash-on-Cash Yield*

- **Cash-on-cash (current) yield for 2006:**

$$y_E = \frac{(7\% - 0.85 \times 4\%)}{(0.15)} = 24\%$$

- **Cash-on-cash yield for 2009: 21.3%**
  - **Lower NOI from leasing property**

⇨ **Problem?**

Japanese Property *– Property Value*

- **Declining property value (as NOI falls)**
- **Assume 7% going out cap rate**

$$V_{2009} = \frac{(6.59 \times (1 - 0.02)}{0.07} = 92.26$$

- **Equity investment left in 2009: 92.26 – 85 = 7.26 (of original 15)**

Japanese Property *– Equity Return from Appreciation*

- **Equity return from capital appreciation:**

$$g_E = \frac{(g_P - LTV \times g_D)}{(1 - LTV)} = \frac{(-7.74\% - 0.85 \times 0)}{(0.15)} = -51.6\%$$

or alternative calculation:

$$g_E = \frac{7.26 - 15}{15} = \frac{-7.74}{15} = -51.6\%$$

- **Annual expected equity capital gain: -16.6% = … *How?***

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Example: Japanese Property — *Continued*

- **Annual expected equity capital gain?**

$$\left(1+r_E^{ann}\right)^4 \times 15 = 7.26 \;\Rightarrow$$

$$r_E^{ann} = \left(\frac{7.26}{15}\right)^{1/4} - 1 = -16.6\%$$

Example: Japanese Property —*Conclusions*

- **Example demonstrates dilemma of Japanese market for property investors**
  - **Attractive cash-on-cash yields from positive leverage**
  - **BUT… declining values!**
- **If economy stabilises or rebounds ⇨ 'Homerun'**
- **If market becomes unexpectedly worse or cap rates rise slightly ⇨ 'Disaster'**

⇨ **<u>Leverage and risk</u> …**

<u>'Focus point 6'</u>

Leverage, cash-on-cash and growth yield – after an example…
Are these concepts clear now?

Leverage and Risk — *Example*

- **Consider RE project**
  - **Total cost (in t=0): 1**
  - **Holding period = 5 years (t=1,…,5)**
  - **Discount rate = 10%**
  - **3 possible scenarios**
    - **Pessimistic:** $NOI_1 = 0.1$; $g = -2\%$; $V_S=0.9$; $p=0.2$
    - **Most likely:** $NOI_1 = 0.11$; $g = 0\%$; $V_S=1.1$; $p=0.5$
    - **Optimistic:** $NOI_1 = 0.12$; $g = +5\%$; $V_S=1.3$; $p=0.3$

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Risk Level for Unlevered Investment

UNLEVERED	(1) Estimated $BTIRR_P$	(2) Expected $BTIRR_P$*	(3) Deviation	(4) Squared Deviation	(5) Probability	(6) Product (4) * (5)
Pessimistic	7.93	13.06	-5.13	26.31	0.2	5.26
Likely	12.56	13.06	-0.50	0.25	0.5	0.12
Optimistic	17.31	13.06	+4.25	18.07	0.3	5.42

*Expected $BTIRR_P$ = (0.2)*7.93 + (0.5)*12.56 + (0.3)*17.31 = 13.06

$\Rightarrow$ Variance $= 5.26 + 0.12 + 5.42 = 10.81$

$\Rightarrow$ Standard Deviation $= \sqrt{10.81} = 3.29$

Risk Level for Levered Investment (Using Fully Amortizing CPM)

LEVERED *(Loan = 0.9, interest rate = 10%, 15-year term, annual compounding)*	(1) Estimated $BTIRR_E$	(2) Expected $BTIRR_E$*	(3) Deviation	(4) Squared Deviation	(5) Probability	(6) Product (4) * (5)
Pessimistic	-6.03	25.46	-31.50	991.95	0.2	198.39
Likely	24.96	25.46	-0.51	0.26	0.5	0.13
Optimistic	47.30	25.46	+21.84	476.98	0.3	143.09

Expected $BTIRR_E$ = (0.2)-6.03 + (0.5)*24.96 + (0.3)*47.30 = 25.46

$\Rightarrow$ Variance $= 198.39 + 0.13 + 143.09 = 341.61$

$\Rightarrow$ Standard Deviation $= \sqrt{341.61} = 18.48$

Formalizing the Relationship between Leverage and Risk

- **Recall** $r = r_f + RP$
- **So applying WACC formula**

  $RP_P = (LTV)\ RP_D + (1 - LTV)\ RP_E$
- **Reformulate to …**

  $RP_E = RP_D + LR\ (RP_P - RP_D)$
- **If debt is riskless then**

  $RP_E = LR \times RP_P$

  *(RP_E is **directly** proportional to LR!)*

Conclusions

⇨ ***Higher risk*** **is implicit cost associated with financial leverage!**

- **No free lunch in the use of leverage!**

⇨ **Always ask…**

- **Is higher expected return commensurate with higher risk?**
- **Ways to realize higher return with less risk?**

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